How-To Library

Informatica

# Generate a GenAI application for vehicle insurance claim processing

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# Abstract

This article helps you use Informatica's Cloud Application Integration's GenAI capabilities to build AI-powered business processes.

# Supported Versions

- Application Integration October 2025

# Table of Contents

Introduction. . . . . 2
Prerequisites. . . . . 3
Traditional vehicle insurance claim processing. . . . . 4
Step 1. Importing assets. . . . . 4
Step 2. Manually assessing the damage and processing the claim. . . . . 5
GenAI-based vehicle claim insurance processing. . . . . 8
Step 1. Create a Gemini connection. . . . . 8
Step 2. Using the LLM to access vehicle damage after an accident. . . . . 10
Create a Google Gemini API key. . . . . 12

# Introduction

Use Informatica's Cloud Application Integration GenAI capabilities to build AI-powered business processes that enhance your efficiency and decision-making. You only need to import pre-built workflows into your environment, and then enhance your existing business processes by adding GenAI features.

**How it works**

Create an intelligent AI-powered application that streamlines insurance claim requests. When you upload vehicle damage through your insurance portal, the embedded large language model (LLM) evaluates the damage and automatically calculates the claim amount. If the LLM can't evaluate the damage, it routes the images to a human agent for manual evaluation through a human task. Meanwhile, the human agent stays informed with timely email notifications throughout the process.

**Benefits of GenAI**

By leveraging these GenAI capabilities, you'll automate complex tasks, reduce manual effort, and improve both accuracy and speed of your business processes. This empowers you to build smarter, more responsive AI-driven processes tailored to your organizational needs.

**Methods to process vehicle insurance claims**

There are two methods to process your vehicle insurance claim:

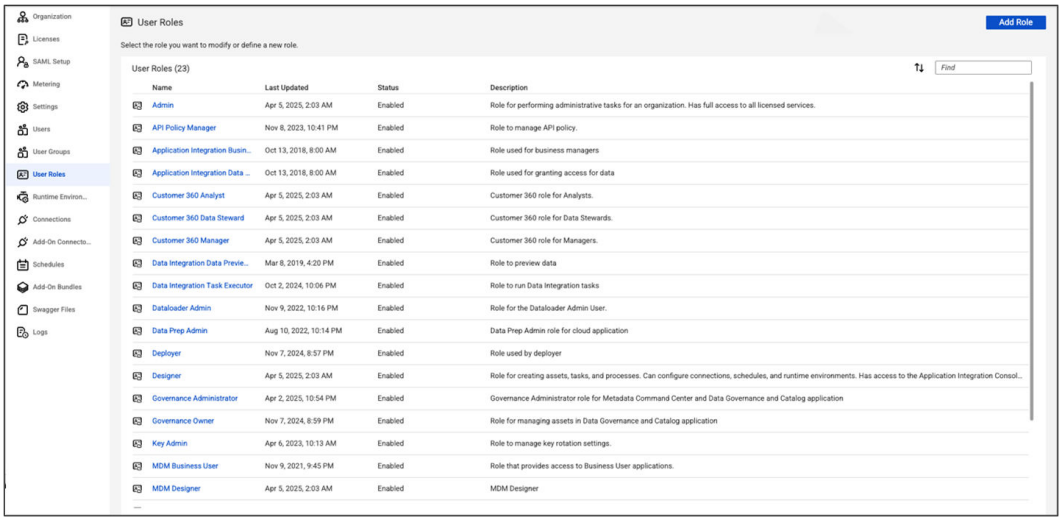
- **Traditional method**: You assign a human agent to manually examine the images, identify the damage, and calculate the payout.
- **Automated method**: You use an LLM through a Google Gemini connection to automatically examine the damage and determine the payout without manual intervention.

## Prerequisites

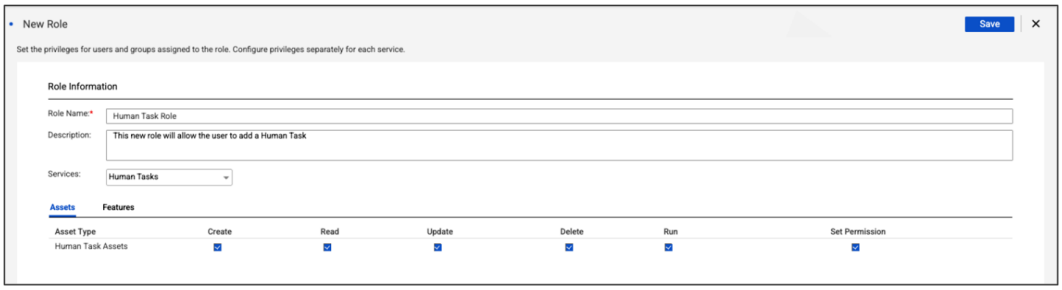
Before you start processing vehicle insurance claims, you must create and assign a Human Task role in your environment. This role enables seamless manual intervention when required during claim processing.

Perform the following steps to create and assign a **Human Task** role:

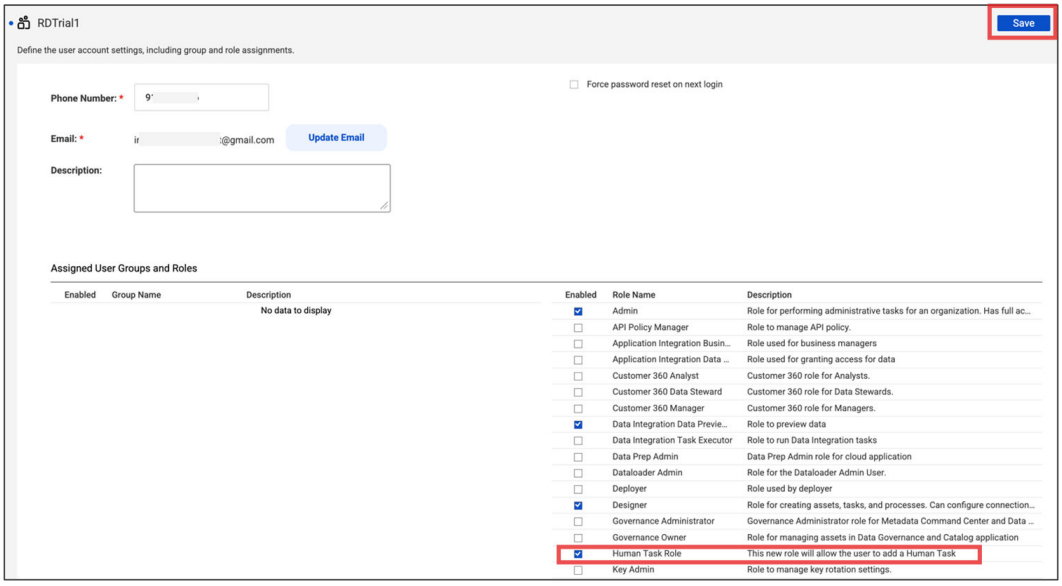
1. Log in to IDMC and click **Administrator**. From the left navigation panel, click **User Roles**.
2. On the **User Roles** page, click **Add Role** on the top right corner.



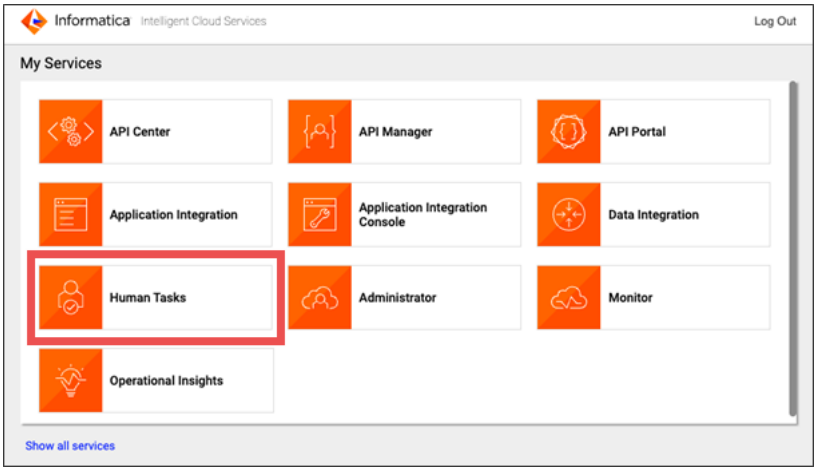

   The **New Role** page appears.
3. On the **New Role** page, enter a name and description for your role.
   For example, let's enter the name as `Human Task Role`.
4. From the **Services** list, select **Human Tasks**.
   Select the check boxes beside the **Create**, **Read**, **Update**, **Delete**, **Run**, and **Set Permission** options in the **Human Task Assets** row.




5. On the **Features** tab, select the check boxes for **Development**, **View Human Task Application**, and **View Tasks**.
6. Click **Save** to create your new role.
7. From the left navigation panel, click **Users**. Click the user name that you used to log into the account.
8. Select the check box next to **Human Task Role** and click **Save**.




9. Log out, and then log back in to IDMC. You'll see the **Human Tasks** service appear on your dashboard, confirming that the role is active.




# Traditional vehicle insurance claim processing

In the traditional method, you review submitted images and supporting documents to identify the damage, diagnoze required repairs, and estimate the payout. You apply policy rules, consult repair guides or vendor estimates, and perform manual data entry and validation. You rely on human judgment to handle complex or ambiguous cases and to make nuanced decisions that require experience. However, this approach tends to be slower and more costly, can vary by examiner, and scales poorly when claim volumes are high.

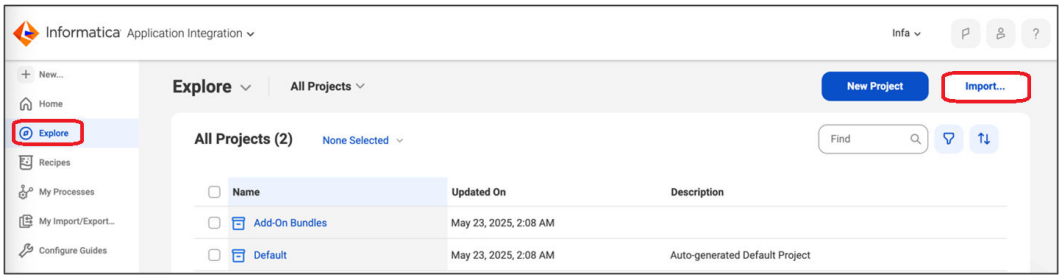
## *Step 1. Importing assets*

Before you manually review images to identify the damaged parts and calculate payouts, you must import and publish the required assets in Application Integration.

Perform the following steps to import the assets:

1. Save the zip file containing the assets to your local system.

2. Log in to IDMC and select **Application Integration**.
3. On the **Explore** page, click **Import** on the top right corner.




4. Click **Choose File** and select the downloaded zip file from your local system.
5. Verify that the following 13 assets appear for import:
   - **ClaimRequest** – Process object that provides the required fields to process the insurance claim
   - **VehicleDetails** – Process object that provides the required vehicle details to process the insurance claim
   - **EmailConnection** – Email connection to send an email
   - **GeminiConnection** – Gemini connection that connects to Google Gemini and performs actions
   - **Check Image Format** – Sub-process that verifies the file type
   - **Calculating the Payout** – Sub-process that calculates the payout for a specified vehicle based on the damaged parts and their prices
   - **Get Price List and Parts List** – Sub-process to get a sample price list and a list of parts for a specified vehicle
   - **Send Email with Claim Details** – Sub-process that sends an email with claim details
   - **Validate Vehicle Information** – Sub-process that checks vehicle details from a sample vehicle dataset
   - **Damage Recognition using LLM** – Sub-process that calls an LLM model to determine the list of damaged parts
   - **Damage Determination** – Human task to determine the damaged parts based on the image of the incident
   - **Process Claim Request** – main process that takes in vehicle information and lets the Human Task calculate the payout, when this process is enabled with an LLM, the LLM calculates the insurance payout for some cases
   - **Insurance Claim Processor** – Guide that demonstrates insurance claim processing with GenAI
6. Confirm that the import location is set to the default location **AI-Powered Vehicle Claim Insurance Processing Google Gemini**.
7. To start importing the assets for the traditional vehicle insurance claim process, click **Import** on the top right corner.

## *Step 2. Manually assessing the damage and processing the claim*

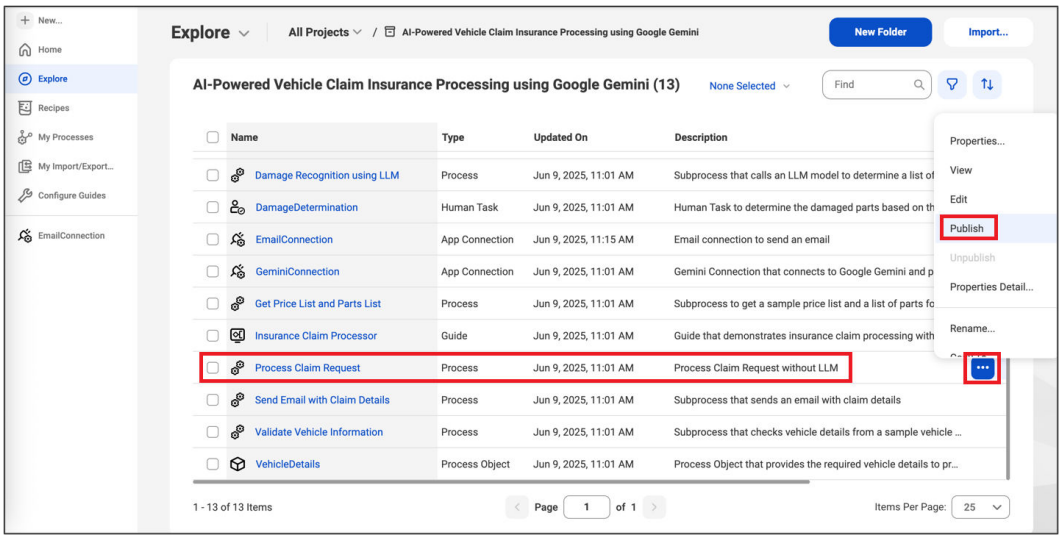
After importing the assets, you must validate and publish them to process claims manually.

Perform the following steps to validate and publish the assets:

1. In Application Integration, navigate to **Explore > AI-Powered Vehicle Claim Insurance Processing using Google Gemini > EmailConnection**.
2. On the **EmailConnection** page, configure the connection by entering the email ID, password, and other details.
3. Save and publish the connection.

**Note:** Use an email service that supports sending emails through third-party applications. For example, the paid versions of Gmail or Outlook.

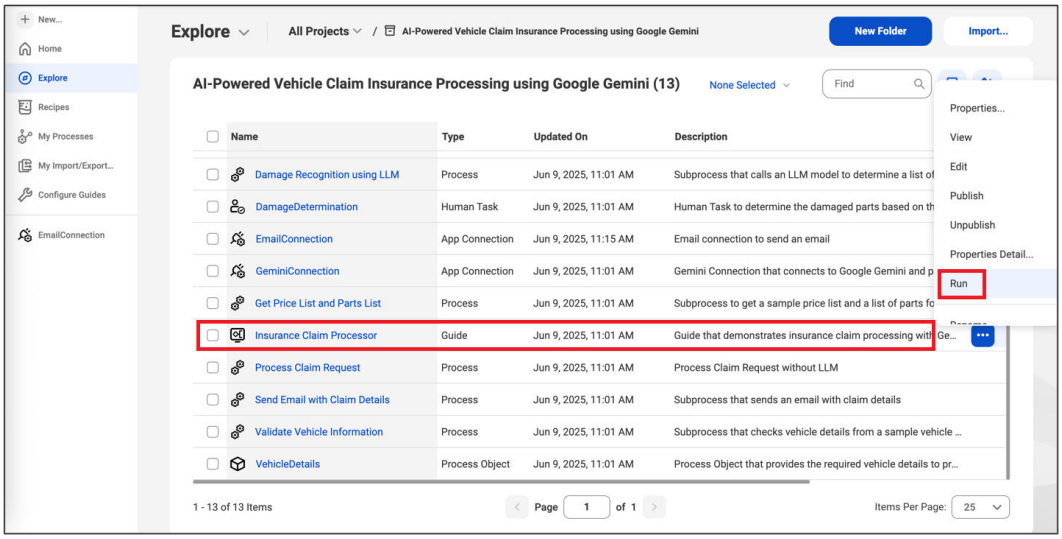
4. Browse to the **Explore** page. Against the **Process Claim Request** process, click the **Actions** menu and click **Publish**.




**Note:** When you publish the **Process Claim Request** process, all its dependent processes are automatically published.

5. Similarly, from the **Explore** page, publish the **Insurance Claim Processor** guide from its **Actions** menu.

6. For the **Insurance Claim Processor** guide, click the **Actions** menu and click **Run**. A new browser tab opens with the guide interface.



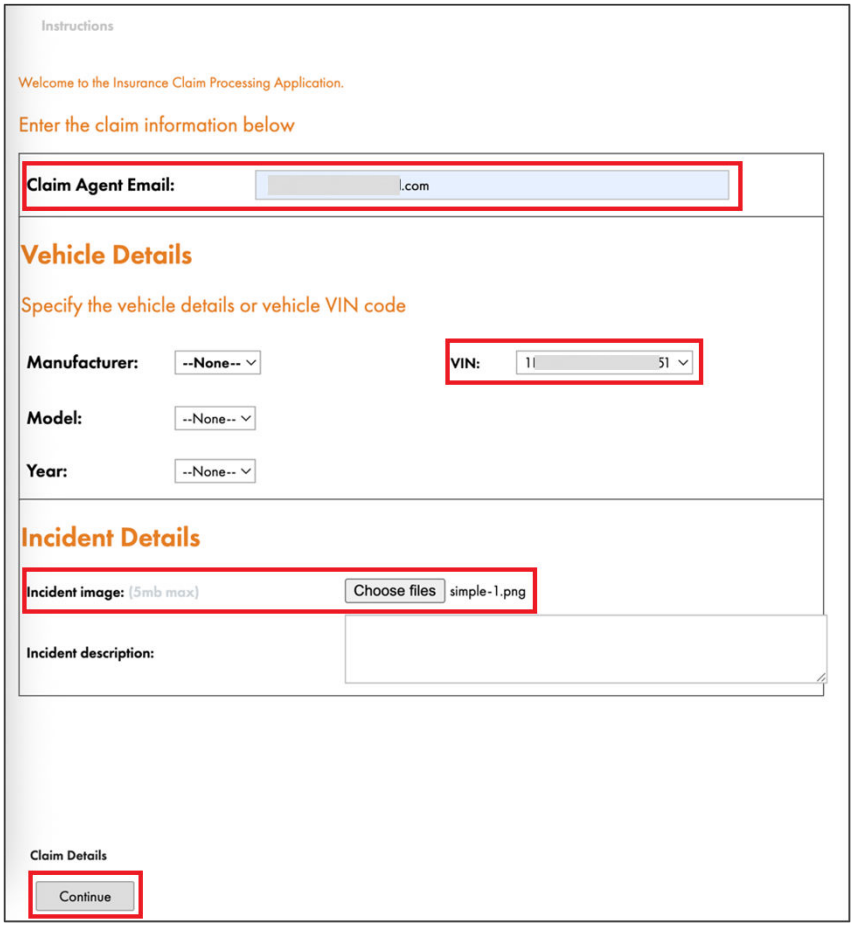

7. In the Insurance Portal, enter the following vehicle owner details:

   a. Enter the email ID to receive claim notification emails. This should be the insurance agent's email ID. For example, enter your own email ID so that you can check the emails later.

      **Note:** This email ID must be different from the one used in the Email connection.

   b. Enter the vehicle details. Instead of manually entering the manufacturer, model, and year, select the **VIN** from the available list.

c. Upload an image of the damaged vehicle and click **Continue**.



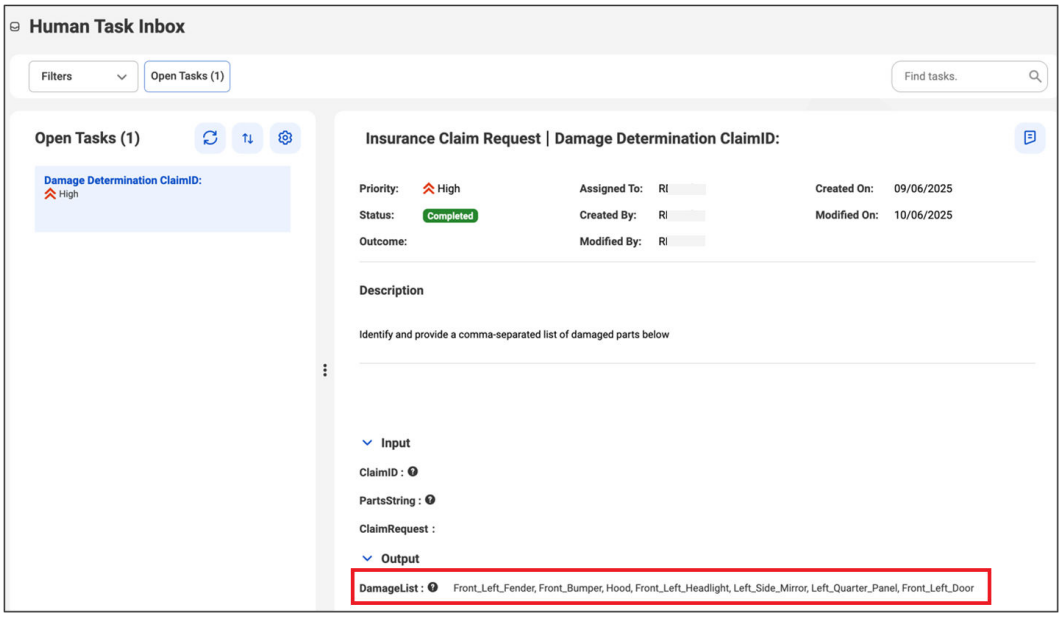

d. The portal generates the **Claim ID** automatically. The insurance agent reviews the claim manually, and identifies the damaged parts.

**Note:** The **ClaimID** field is auto-generated and changes every time you run the process.

e. Click **Continue > Done**.
You receive an email including the vehicle image and a master list of parts eligible for damage claims. Check your spam folder if you don't receive the email.

8. Log in to IDMC, and select **Human Tasks**, and refresh your inbox. Select the new task and click **Start**.

9. Refer to the vehicle image in the email, enter the list of damaged parts based on the image, and click **Complete**.
   The claimed damaged parts appear on the screen.




You will receive an email confirming the damages and the final payout amount that you are eligible to receive.

This method highlights the traditional insurance claim processing method, where a human agent manually evaluates damage in every claim.

# GenAI-based vehicle claim insurance processing

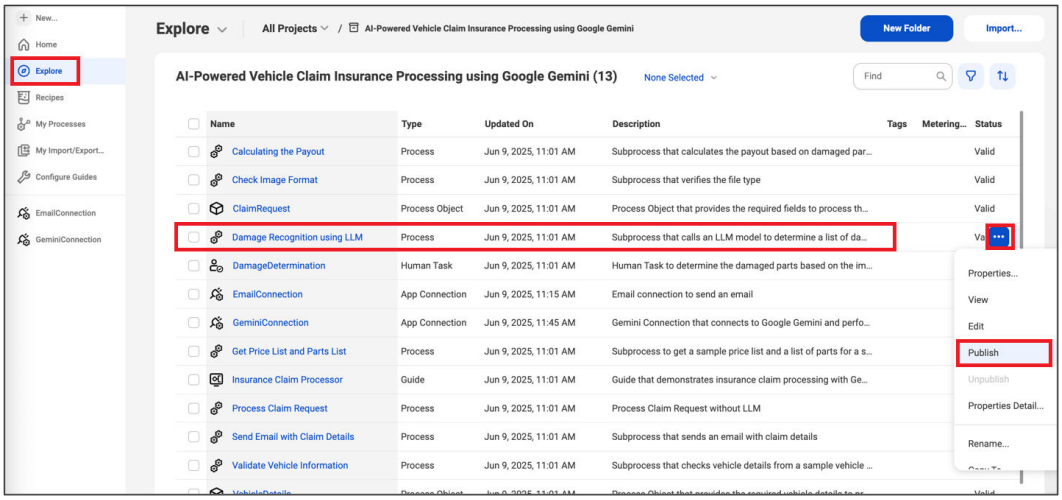
In the automated GenAI-based method, you route images and metadata to an LLM through a Google Gemini connection to analyze damage, classify severity, and map findings to rule-based payout calculations. The LLM produces a structured assessment, suggests repair items, and recommends a payout that integrates with downstream workflows for approval and payment. This method increases processing speed, reduces per-claim cost, and delivers consistent initial decisions at scale. It requires careful model validation, robust exception handling, strong data-privacy safeguards, and human review for edge cases or high-value claims.

## *Step 1. Create a Gemini connection*

Use Informatica's Application Integration GenAI capabilities to create an intelligent AI-powered process that automatically evaluates insurance claims, calculates payouts, and notifies insurance agents by email. By adding an LLM to your process, you reduce manual work. If the LLM cannot accurately assess damage, the process redirects the claim to a human agent for evaluation.

Perform the following steps to add an LLM to a process:

1. In **Application Integration**, click **Explore** and select **GeminiConnection** from the asset list.
2. Enter the Gemini API key. Click **Save** and **Publish**.
   For more information about generating a Gemini API key, see Create a Google Gemini API Keyon page 12.
3. Browse to the **Explore** page. Against the **Damage Recognition using LLM** process, click the **Actions** menu and click **Publish**.

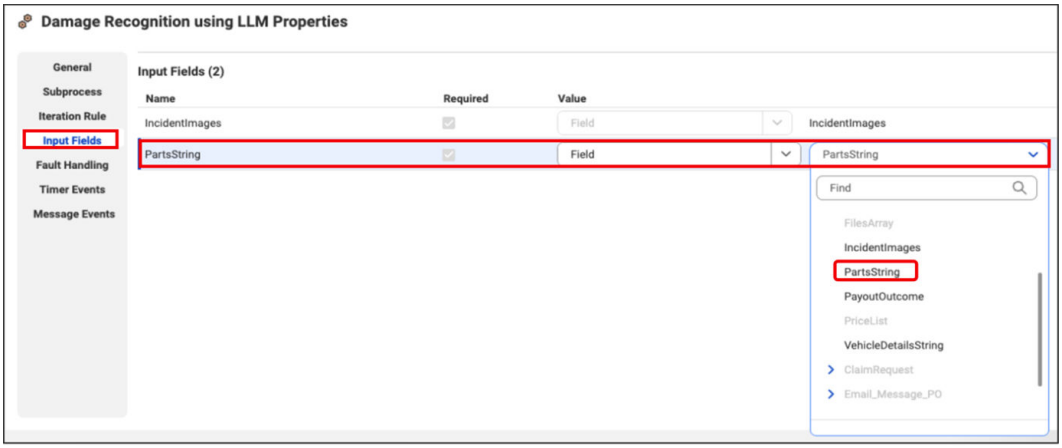



4. Open the **Process Claim Request** process. Add a subprocess named **Damage Recognition using LLM** between the **Check Image Format** subprocess and the **ExtentOfDamage** assigment step.
5. On the **Subprocess** tab, select **Damage Recognition using LLM** in the **Process Call** field.




The remaining field values will auto populate.

6. Enter the following values on the **Input Fields** tab:
   - For the first input row, select **Field** and choose **IncidentImages** as the field name.
   - For the second input row, select **Field** and choose **PartsString** as the field name.

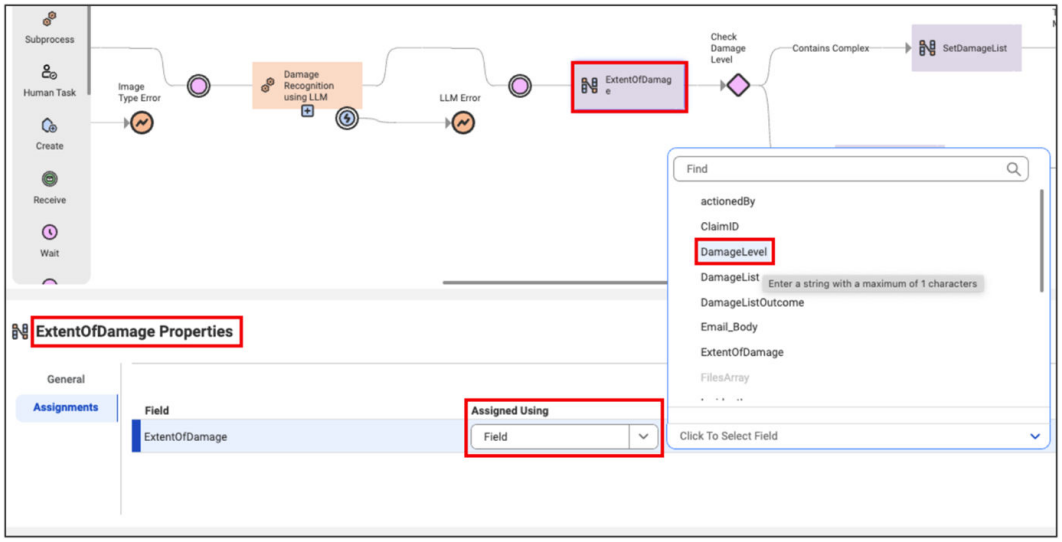



7. Enable **Catch Faults** on the **Fault Handling** tab.

8. Add a **Throw** event for the **Damage Recognition using LLM** subprocess in the Process Designer. Name this event as **LLM Error** in the **Properties** section of the **General** tab.
   On the **Fault Fields** tab, set values for the following fields:

| Field Name | Value |
|---|---|
| Code | Field. faultinfo>code |
| Detail | Field. faultinfo>detail |
| Reason | Field. faultinfo>reason |

9. On the **Assignments** tab of the **ExtentOfDamage** step, set **Assigned Using** to **Field**, and **From** to **DamageLevel**.




10. Save and publish the process.

You have now added the LLM subprocess to the process. Your process can now send images to the LLM for automatic damage assessment and insurance payout calculation.

## *Step 2. Using the LLM to access vehicle damage after an accident*

After you import and configure the assets, validate, publish, and run the **Insurance Claim Processor** guide to process claims.

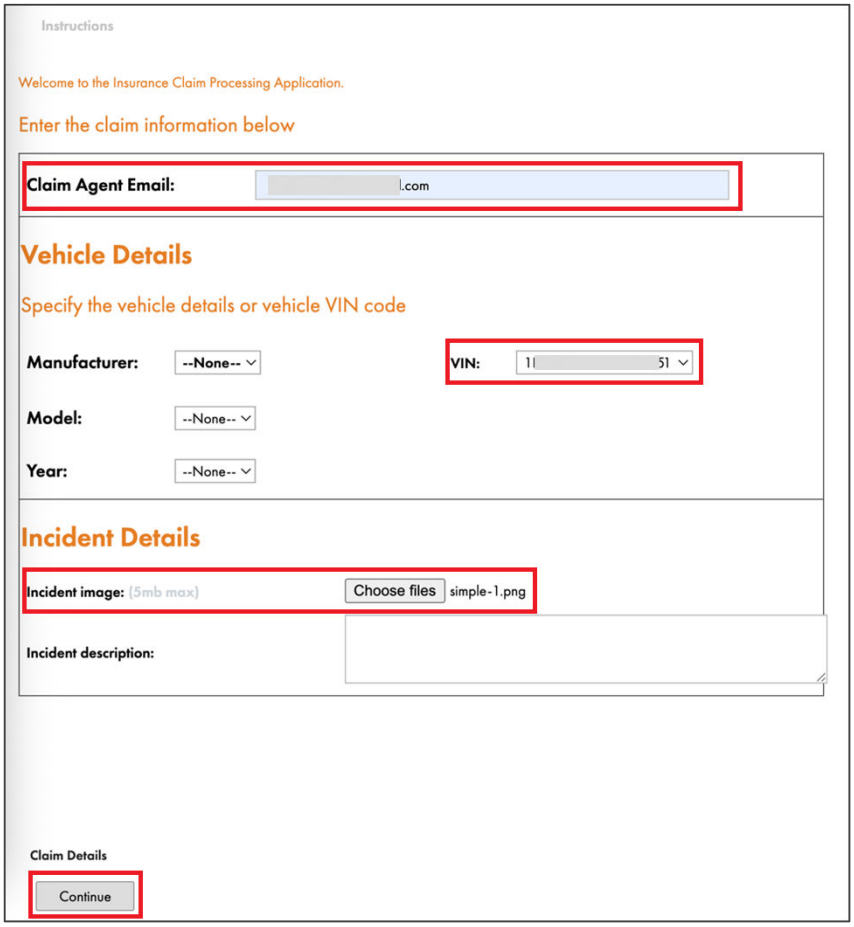
Perform the following steps to validate and publish the imported assets, and to run the guide:

1. In Application Integration, browse to the **Explore** page. Against the **Insurance Claim Processor** guide, click the **Actions** menu and click **Run**.
   A new browser tab opens with the insurance portal.
2. Enter the following details:
   a. Enter the email ID to receive claim-related emails. This should be the insurance agent's email ID.

      **Note:** This email ID must be different from the one used in the Email connection.

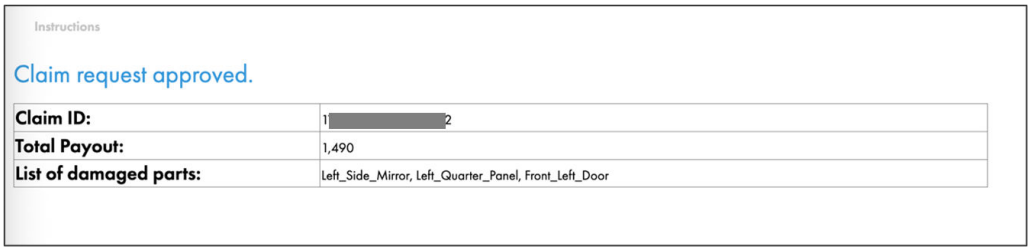
   b. Enter the vehicle details by selecting the **VIN** from the available list instead of manually entering the manufacturer, model, and year.

c. Upload an image of the damaged vehicle and click **Continue**.




3. Click **Continue > Done**.

- If the LLM successfully determines the damage, you will see an on-screen message with the claim details. You will also receive an email confirming this information.

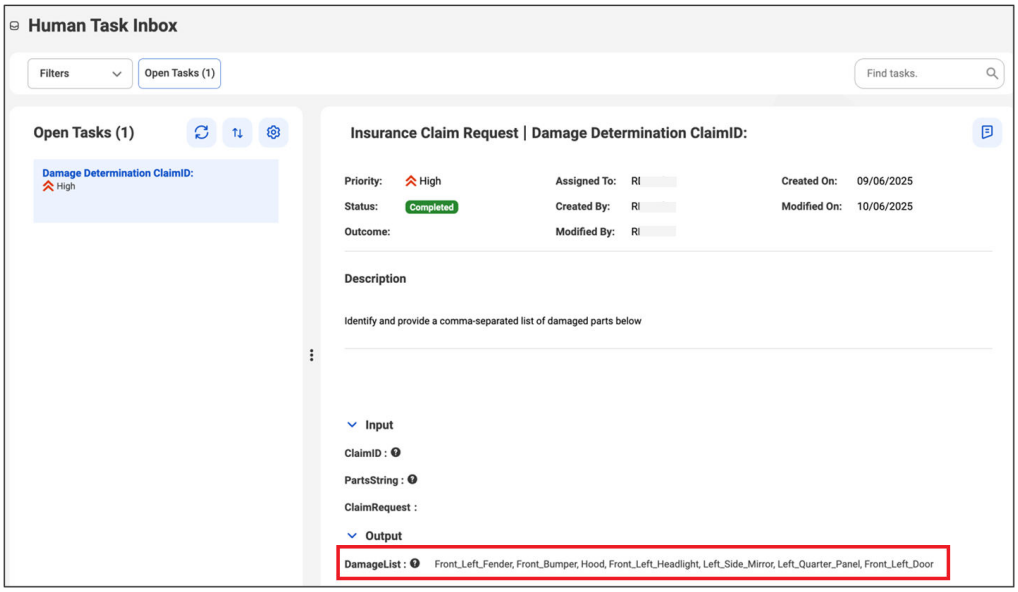



- If the LLM can't determine the damage, the process redirects to human agents through **Human Tasks** for manual evaluation.

Perform the following steps to complete the human assessment if needed:

1. Log in to IDMC, and select **Human Tasks**, and refresh your inbox. Select the new task and click **Start**.

2. Refer to the vehicle image in the email, enter the list of damaged parts based on the image, and click **Complete**.
   The claimed damaged parts appear on the screen.



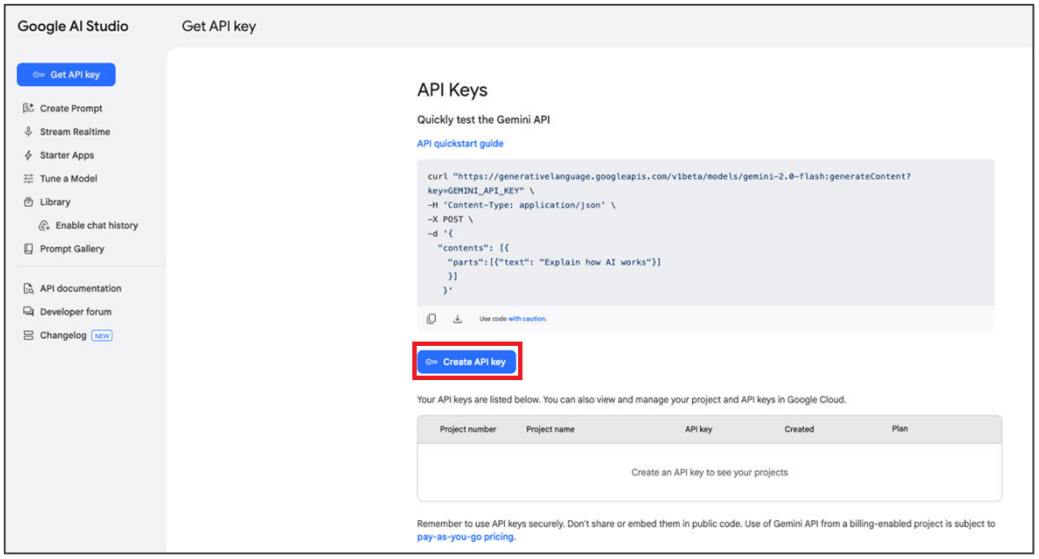

You will receive an email listing the damaged parts and the final payout amount that you are eligible for.

This method shows how the LLM automates damage assessment and highlights where human intervention is necessary.

## *Create a Google Gemini API key*

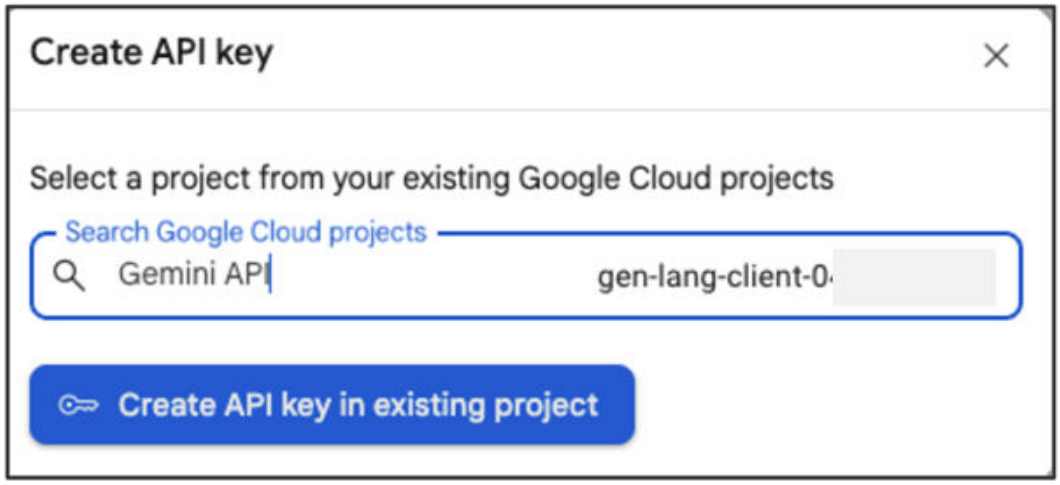
Perform the following steps to receive a Google Gemini API key from your personal Google or Gmail account:

1. Log in to your Google or Gmail account. Open https://aistudio.google.com/app/apikey on a new tab.
2. Click **Create API key**. Accept the terms and conditions if prompted.

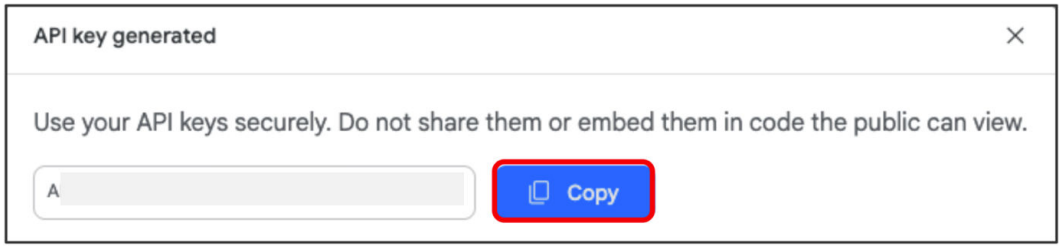



An API key is created.

3. If prompted, select a project and click **Create API key in existing project**.




4. Copy the created API key and use it as required in your Gemini connection set-up.




# Author

**Informatica Documentation Team**